1. In the expression Y + X'. Y, which operator will be evaluated first?
   (a) ' (b) + (c) . (d) ,

Computer – Boolean Algebra (Operator Precedence) – 2021

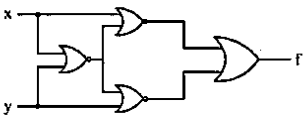
2. Which of the following is false?
   (a) x + y = y + x (b) x.y = y.x
   (c) x.x'= 1 (d) x + x' = 1

Computer – Boolean Algebra Laws – 2021

3. A Boolean function x'y' + xy + x'y is equivalent to
   (a) x' +y' (b) x + y
   (c) x + y' (d) x' + y

Computer – Boolean Expression Simplification – 2021

4. Which of the following logic operations is performed by the following given combinational circuit?



   (a) Exclusive-OR (b) Exclusive-NOR
   (c) NAND (d) NOR

Computer – Logic Gates – 2021

5. The output of an AND gate with three inputs A, B and C is HIGH when
   (a) A = 1, B = 1, C = 0 (b) A = 0, B = 0, C = 0
   (c) A = 1, B = 1, C = 1 (d) A = 1, B = 0, C = 1

6. Don't care conditions can be used for simplifying Boolean expressions in
   (a) registers (b) terms
   (c) K-maps (d) latches

Computer – Karnaugh Map – 2021

7. In the expression A + BC, the total numbers of minterms will be
   (a) 2 (b) 3
   (c) 4 (d) 5

Computer – Boolean Minterms – 2021

8. Given the following truth table:

A	B	x
0	0	1
0	1	0
1	0	0
1	1	0

   Which of the following Boolean functions does it represent?
   (a) OR (b) XOR
   (c) NOR (d) XNOR

COMPUTER – Boolean Logic– 2022

9. The simplified form of Boolean expression AB + AB' is ____________.
   (a) A (b) B (c) 1 + A (d) 1 + B'

COMPUTER – Boolean Algebra– 2022

10. Match List I with List II

	List I		List II
A.	$(x+y)'$	I.	I
B.	$x+1$	II.	$x'+y'$
C.	$(xy)'$	III	$x'.y'$
D.	$x+0$	IV.	x

   Choose the correct answer from the options given below:
   (a) A – II; B – I; C – III; D – IV
   (b) A – II; B – IV; C – I; D – III
   (c) A – III; B – I; C – IV; D – II
   (d) A – III; B – I; C – II; D – IV

COMPUTER – Boolean Identities– 2022

11. The simplified form of Boolean Expression:
    $\bar{A}BC$ + ABC is ____________.
    (a) AC (b) AB
    (c) BC (d) 0

COMPUTER – Boolean Algebra– 2022

12. If ⊕ and ⊙ denote the exclusive OR and exclusive NOR operations, respectively, then which one of the following is not correct?
    (a) $\overline{P\oplus Q} = P \odot Q$
    (b) $\bar{P}\oplus Q = P \odot Q$
    (c) $\bar{P}\oplus\bar{Q} = \bar{P}\oplus Q$
    (d) $(P\oplus\bar{P}) + Q = (P \odot \bar{P}) \odot \bar{Q}$

COMPUTER – Boolean Algebra **– 2023**

13. If x, w, y, z are Boolean variables, then which of the following is incorrect?
    (a) $.wx + w(x+y) + x(x+y) = x + wy$
    (b) $\overline{w\bar{x}(y+\bar{z})} + \bar{w}x = \bar{w} + x + \bar{y}z$
    $(c)(w\bar{x}(y+x\bar{z}) + \bar{w}\bar{x})y = x\bar{y}$
    (d)$(w+y)(wxy+wyz) = wxy + wyz$

COMPUTER – Boolean Algebra **– 2023**

14. The number of possible Boolean function that can be defined for n Boolean variables over n- valued Boolean algebra is _______.
    (a). n^{2^n} (b). 2^{n^2}
    (c). 2^{2^n} (d). n^{n^n}

COMPUTER – Boolean Function **– 2023**

15. The function represented by the k-map given below is

k-map				
C/AB	00	01	11	10
0	0	1	1	1
1	0	0	1	0

    (a) BC + AB + AC' (b) BC' + AB + AC'
    (c) (B ⊕ C)' (d) A.BC

Digital Logic – K-Map Simplification

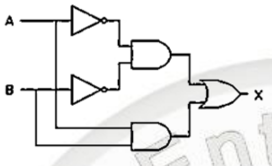
16. Let A’ represent complement of A. Which of the following Boolean expression is/are true?
(A) A + AB = A (B) (A + B)’ = A’B’
(C) (A')’ = A (D) (AB)’ = A’ + B’
Choose the correct answer from the options given below:
(a) A, B and D only (b) A and D only
(c) A, B, C and D (d) B and D only

Boolean Algebra – Laws & Identities

17. The following circuit generates the same ouput as?

(CUET-PG 2025) Logic Gate

(a)XOR GATE
(b)XNOR GATE
(c)NAND GATE
(d)NOR GATE



18. Consider the following Karnaugh Map (k-map). Minimal Function generated by this Kamaugh map is

RS \ PQ	00	01	11	10
00	1	1		1
01	X			
11	X			
10	1	1		X

(CUET-PG 2025) K- map

(a) Q'S + P'Q (b) Q'S' + P'S'
(c) P'Q' + P'S' + Q'S' (d) PQ + Q'S' + P'S'

19. What should be the output of the following boolean expression after simplifying it to a minimum number of variables?
a'b' + ab + a'b

(CUET-PG 2025) Boolean Algebra

(a) a + b' (b) a' + b
(c) a' + b' (d) a + b

ANSWER KEY

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
a	c	d	a	c	c	d	c	a	d
11.	12.	13.	14.	15.	16.	17.	18.	19.	
c	d	c	c	b	c	b	b	b	